

Chapter 115 — Electrical Stimulation and Neuromodulation in Storage and Emptying Failure

HISTORY OF ELECTRICAL STIMULATION

- Electrical stimulation of the lower urinary tract (LUT) innervation is **successful and multipotent**; the idea of evoking a specific organ response through the nervous system is old and **not confined to urology**.
- **Scribonius Largus (CE 46), *Compositiones medicae***: live torpedo fish (also ray and eel) for headache and gout. For podagra the patient stood on a wave-washed shore on a black torpedo fish until numb to the knee — relieving the present attack and preventing recurrence.
- **Battery (18th century)** → new medical uses of electricity; **transistors (1950s)** → first implanted pacemakers; cardiac success → rapid spread to urology.

Muscle-based era

- **Aim: contract the bladder** (for poor contractility) **or the sphincter** (for stress incontinence).
- Sequence: direct detrusor stimulation (Bradley, 1964) → first urethral sphincter stimulator (Caldwell, 1965) → stimulation of sacral cord nuclei in the conus medullaris to induce voiding (Nashold, 1972) → surface, deep and cuff electrodes (Jonas, 1976).
- Canine work failed because **bladder and sphincter were activated simultaneously**.
- Human trials followed in **paraplegia and spina bifida** (impaired sensation made stimulation tolerable), yet direct stimulation proved **painful and only locally effective**.
- **Indirect stimulation via afferent nerves** needed less current and treated the whole end organ — the start of practical clinical neurostimulation.

Brindley's sacral anterior root stimulator

- **Sacral anterior root stimulator (SARS), Giles Brindley**: stimulates the anterior sacral roots → bladder contraction (voiding), defecation and penile erection.
- Anterior stimulation **cannot reduce detrusor overactivity (DO)** → combined with **sacral dorsal rhizotomy** of the same roots; over 25 years this markedly improved quality of life (QoL) in paraplegic patients. Further development by Sauerwein (Bad Wildungen) and Sarrias (Barcelona).

Sacral neuromodulation

- **About 1985: sacral neuromodulation (SNM), or sacral nerve stimulation (SNS)** — sacral root stimulation could **decrease** rather than increase bladder activity. A breakthrough for overactive bladder (OAB), which previously had no effective treatment. Pioneers: **Emil Tanagho and Rick Schmidt**; priority of discovery unclear.
- **Functional magnetic resonance imaging (fMRI)** showed the mechanism is indirect: sacral roots are mixed → current excites **afferents** → **central nervous system (CNS)** → **efferents to pelvic organs**. The whole storage-micturition system is influenced, not the organ directly — hence **"neuromodulation"**.

Tibial and newer targets

- **Percutaneous tibial nerve stimulation (PTNS)**, about 2000: derived from acupuncture point **SP6** at the ankle; **McGuire (1983)** first stimulated the posterior tibial nerve for bladder control; widespread use in the 2000s.
- PTNS is **simpler and cheaper than SNM**; with **no implant and discontinuous stimulation**, a sham-controlled net effect could be convincingly shown.

- Newer targets: dorsal genital nerve, pudendal nerve, intravaginal, **transcutaneous tibial nerve stimulation (TTNS)** by patch; small rechargeable devices entering practice.

Concept — *Direct organ stimulation failed on **pain and locality**; every durable technique since (SARS with rhizotomy aside) works **indirectly through afferents**, which is why the field is neuromodulation rather than muscle stimulation.*

PATHOPHYSIOLOGY OF STORAGE AND VOIDING DISORDERS

- **Exclude anatomical causes before any electrical therapy:** external urethral sphincter (EUS) weakness, poorly compliant detrusor, bladder stones, bladder cancer, vesicovaginal fistula.
- How electrical therapy helps functional disorders is unknown; **afferent modulation of central pathways** probably dominates.

Pathophysiology of Storage Disorders

- Storage failure arises from the **bladder wall** (urothelium, detrusor), the **outlet**, or both. With known neurological disease → **neurogenic lower urinary tract dysfunction (LUTD)**. Electrical stimulation is **mainly approved for non-neurogenic** disorders.
- Continence = **synergic antagonism of bladder and outlet**, EUS the key structure. Filling: bladder relaxed, **EUS tonically contracted — involuntary, cannot be voluntarily interrupted**.
- **Afferents:** bladder filling via **Aδ fibres** in the pelvic and hypogastric nerves (from urothelium, suburothelium, detrusor); bladder neck and urethra via **pudendal and hypogastric** nerves → lumbosacral cord (lateral dorsal horn, intermediate zone) → mostly intraspinal proprioceptive connections, some ascending to pons and midbrain; the **spinothalamic tract** keeps the sensory cortex informed of fullness.
- **OAB:** urge cannot be deferred but can be **suppressed** — pelvic floor physiotherapy, drugs, botulinum toxin, electrical stimulation.
- **Sympathetic outflow fires only from ≈60% of capacity** → wall relaxation and smooth-muscle closure of the bladder neck. Sympatholytics or sympathectomy barely affect storage → the **sympathetic system is not the main continence mechanism** in health.

Pathophysiology of Voiding Disorders

- **Exclude anatomical obstruction first:** urethral stricture or stone, benign prostatic enlargement, pelvic organ prolapse.
- **Detrusor cause:** underactive detrusor = **hypocontractility plus hyposensitivity**. **Outlet cause:** anatomical obstruction, or neurogenic **detrusor-external sphincter dyssynergia (DESD)**.
- Electrical stimulation **cannot restore muscle volume or correct anatomical obstruction**.
- Normal voiding sequence: **EUS relaxes → 1–2 s later detrusor contracts**. Without EUS relaxation, voiding is impossible.
- Circuit sequence from the **pontine micturition centre (PMC)**: long axons → parasympathetic preganglionic motoneurons (sacral intermediolateral column) → postganglionic neurons in bladder wall → **delayed smooth-muscle detrusor contraction**; collaterals → inhibitory **γ-aminobutyric acid (GABA)/glycine interneurons** (sacral intermediomedial column) → EUS motoneurons → **immediate striated EUS relaxation**. Both arms fire simultaneously at sacral level.
- **Forebrain operates the PMC as a switch:** forebrain lesions lose the **timing** of micturition but the **reflex remains intact**. Positron emission tomography (PET) and fMRI confirm the PMC in humans. Afferents → thalamus → **cingulate and insular cortex** → **prefrontal cortex** decides whether voiding is safe.

- **Kegel exercises** add voluntary motor-cortex contraction on top of tonic involuntary contraction; on stopping, tone returns to baseline — so **physiotherapy is usually ineffective in retention**.

PONTINE STORAGE CENTRE VERSUS PONTINE MICTURITION CENTRE

	Pontine storage centre (PSC)	Pontine micturition centre (PMC)
Alias	L-region	M-region
Site	Lateral pontine tegmentum	Pontine tegmentum, medial to PSC
Output	Tonic involuntary EUS contraction	EUS relaxation (indirect, inhibitory interneurons) then detrusor contraction (direct)
Lesion (cat)	Bilateral ablation → cannot store	Final common pathway: no switch-on → no voiding

Concept — Storage is **tonic and involuntary**, voiding is a **switched event gated by EUS relaxation**; electrical therapy can act only on this neural control, never on anatomy or muscle bulk, which is why anatomical causes are excluded first.

WORKING MECHANISM OF ELECTRICAL STIMULATION OF THE LOWER URINARY TRACT

- Electrical therapies act not only on the LUT and spinal cord but on **supraspinal centres**, which in turn modulate the LUT.

Storage Disorders

- No mechanism is confirmed. Theories concern SNS but probably **apply to all electrical therapies**.
- **SNS differs in scale**: it recruits **≥1000 times more axons, continuously**, versus a small or superficial nerve stimulated intermittently; the implant also gives more flexible parameters.
- A common central mode may extend to faecal incontinence, chronic pain and deep brain stimulation.
- Storage disorders are attributed to **loss of inhibitory control acquired during postnatal maturation** of the voiding reflex; stimulation may **restore** it.

THEORIES OF THE MECHANISM OF SACRAL NERVE STIMULATION (FIGS. 115.1–115.4)

Theory	Proposed pathway	Status
Early (efferent)	Efferent motor nerves → pelvic muscle contraction → detrusor relaxation via intramural postganglionic cells	Less plausible : benefit needs no pelvic floor contraction; acute SNS leaves urodynamics unchanged; pure sensory nerves (dorsal clitoral, dorsal penile) give similar effect
Spinal cord (2005)	Afferents inhibit sacral interneurons and preganglionic parasympathetic motoneurons	Superseded
Spinal cord–PMC	SNS inhibits PMC activation → fewer OAB contractions	Alternative to above
Forebrain (current)	Afferents → forebrain areas of awareness and alertness → less PMC firing	Current view ; PET shows regional cerebral blood-flow change in sensory and attention areas; chronic SNS → neuroplasticity

Voiding Disorders

- Paradox: SNS treats **both overactive and underactive bladder**.
- Functional voiding disorders lack effective filling sensation and **cannot switch on the PMC**. SNS raises **awareness of filling and of pelvic floor tone** → PMC switched on → EUS relaxes.

- In underactive bladder (UAB) SNS works **mainly by striated sphincter relaxation**, much less by detrusor contraction.

Concept — *One afferent → forebrain mechanism serves opposite disorders: in OAB heightened control **damps PMC firing**; in UAB heightened awareness **lets the PMC switch on**.*

INTRAVAGINAL ELECTRIC STIMULATION

- Alternative conservative therapy for **stress urinary incontinence (SUI)**: non-implantable vaginal electrodes **passively activate the pelvic floor directly**, perhaps also indirectly via the pudendal nerve; given by physiotherapists or at home.
- Systematic reviews: **minimally invasive and safe**, but evidence too low in quality to judge effectiveness; one review found better pad use and subjective outcome. All call for higher-quality trials.

TRANSCUTANEOUS SACRAL DERMATOME STIMULATION

- **Transcutaneous sacral dermatome stimulation (TSDS)**, established in pain management: current passes skin receptors → posterior root ganglion → spinal centres via **Aβ, C and Aδ fibres**; applied to suppress DO in OAB.
- **Randomised trial, bilateral S3 dermatome electrodes**, refractory idiopathic DO (IDO) and neurogenic DO (NDO): **higher cystometric capacity**, later first involuntary contraction (significant in IDO only), smaller neurogenic contractions.
- **Multiple sclerosis (MS) with NDO, conditional stimulation** (triggered when detrusor pressure exceeded 10 cmH₂O): dorsal genital nerve stimulation suppressed contractions; sacral dermatome stimulation **failed in every patient**.
- **Children** with OAB or dysfunctional voiding: daily S3 stimulation for 3 months (20 min, 10 Hz, 350 μs) → less incontinence, fewer urinary tract infections (UTIs), lower post-void residual (PVR).
- Synthesis: can raise capacity and perhaps lower maximal detrusor pressure (**renal protection**), but is **not useful for conditional stimulation**; studies are few and small.

Concept — *Dermatome stimulation **modulates chronically but cannot abort a contraction acutely**; that gap is what the purely sensory dorsal genital nerve fills.*

PUDENDAL NERVE STIMULATION

- **Anatomy**: S2–S4 → over the ischial spine → **pudendal (Alcock's) canal**; gives the inferior rectal nerve proximally, then the perineal nerve and dorsal genital nerve distally.
- **Low-frequency afferent stimulation inhibits OAB**; dorsal genital stimulation in complete thoracic or cervical spinal cord injury (SCI) strongly inhibits the micturition reflex.
- **Technique**: percutaneous via the pudendal canal with an **off-label tined quadripolar lead**; **posterior approach preferred** — perineal approach carries more lead infection.
- **Indications**: IDO, NDO, chronic pelvic pain.
- **Versus SNM** (blinded crossover, OAB and non-obstructive urinary retention [NOUR]): pudendal gave **greater symptom reduction**, and most patients chose it for the permanent implant. Also effective after **failed SNM** → a salvage option.
- **Not approved by the US Food and Drug Administration (FDA)** for chronic stimulation in voiding dysfunction or pelvic pain.

DORSAL GENITAL NERVE STIMULATION

- **Dorsal genital nerve stimulation (DGNS)** — dorsal clitoral or dorsal penile nerve, needle electrodes — is **experimental** for OAB from IDO and NDO; not in routine practice.
- **Theoretical advantages:** easily reached by needle (like the tibial nerve); a **direct pudendal branch**, so a short route to the sacral LUT control; **purely sensory**, so theoretically best at carrying the stimulus afferently to the brain.
- **Correct stimulation confirmed by the genitoanal reflex.**
- **Evidence:** acute DO suppression in SCI; short-term capacity increase in NDO, consistent across small studies (meta-analysis); in women with OAB-wet, 7 days of continuous stimulation reduced incontinence and severe urgency; **on-demand** stimulation (pilot) aims to avoid **habituation** and save energy.
- **Drawbacks:** hard to keep electrodes in contact with the small branches; the stimulation site (dorsal penile base, clitoral area) is problematic for patients. Evidence is mostly acute and small.

Concept — *If neuromodulation works through afferents, the **purest afferent target** should be the most efficient — DGNS is the test of that logic, limited by **fixation and site** rather than by physiology.*

TRANSCUTANEOUS TIBIAL NERVE STIMULATION

- **PTNS:** introduced 1999, FDA approval 2000 (office-based OAB treatment), **National Institute for Health and Clinical Excellence (NICE) approval 2006**. Standard: **12 sessions × 30 min**, 1–3 times a week, plus maintenance → heavy burden on patients and clinics.
- **TTNS** (table below) cuts travel and cost, but the **skin acts as a capacitor**: higher, often painful amplitude is needed, and lowering it for comfort gives suboptimal stimulation.
- **Systematic review** (randomised controlled trials [RCTs] and cohorts; idiopathic and neurogenic OAB — MS, Parkinson disease, stroke, SCI): heterogeneous, underpowered, non-standardised dosing (4–12 weeks; 20–30-min sessions; daily to weekly). Meta-analysis: clinically and statistically significant fall in the **International Consultation on Incontinence Questionnaire–Urinary Incontinence Short Form (ICIQ–UI–SF)** in older adults; **no adverse events**. Verdict: potentially effective in idiopathic OAB, safe in neurogenic OAB; no head-to-head PTNS data.
- **Later review of RCTs:** TTNS as effective as pelvic floor muscle training (PFMT) and PTNS, less effective than antimuscarinics; low cost and self-management make it an attractive alternative. Place in the OAB algorithm unsettled.

PERCUTANEOUS VERSUS TRANSCUTANEOUS TIBIAL NERVE STIMULATION

	PTNS	TTNS
Electrode	34-gauge needle + surface ground pad	Surface electrodes only
Setting	Outpatient clinic, 12 weekly visits, then maintenance	Home, self-administered , patient-set schedule
Efficacy	More effective (bypasses skin)	Less effective: amplitude limited by pain

Concept — *Tibial therapy trades **efficacy against burden**: the needle beats the skin, but the clinic schedule drives both TTNS and the implantable devices.*

PERCUTANEOUS TIBIAL NERVE STIMULATION

- SNS is the **most documented** neuromodulation for OAB; PTNS is the extensively documented alternative, **cheaper overall**, hence wider reach.

- **Anatomy (Fig. 115.9):** LUT innervation **L2–S4**; sciatic nerve **L4–S3** → posterior tibial nerve, whose fibres arise mainly from **S2–S3**.
- **Proposed mechanism:** "crosstalk" between sympathetic and parasympathetic postganglionic terminals at neuroeffector junctions alters efferent signalling to the bladder; peripheral nerves (pudendal, dorsal genital, tibial) can thus modulate bladder behaviour.
- **Animal model (cat, intravesical acetic acid = OAB model):** naloxone blocked PTNS inhibition of the OAB-model DO but **not** of normal reflex activity → **opioid receptors** involved.
- **Somatosensory evoked potentials (SEPs):** after PTNS versus sham, long-latency SEP (P80, P100) amplitude changed → lasting change in synaptic efficiency (**long-term potentiation or depression**) → long-term effect of PTNS and SNS may reflect **cortical network reorganisation**.

AFFERENT FIBRES: OVERACTIVE VERSUS NORMAL REFLEX BLADDER ACTIVITY

	OAB-type reflex activity	Normal reflex activity
Afferent fibre	Nociceptive C fibres	Aδ fibres
PTNS inhibition blocked by naloxone?	Yes (opioid-mediated)	No

HISTORY OF PERIPHERAL NEUROSTIMULATION

- Derived from **acupuncture**; **Wilhelmus Ten Rhyne (1683)**, Dutch physician, described it in Europe (*Dissertatio de Arthritide ... De Acupunctura*).
- **SP-6 (SanYin Jiao):** medial lower leg **≈6–8 cm above the medial malleolus**, near the medial tibial crest; site and target organs closely match PTNS.
- **Electroacupuncture at 2–15 Hz, 10–20 mA resembles PTNS**; difference: PTNS uses an **anatomical substrate**, acupuncture "energetic pathways".
- **McGuire (1983):** first tibial stimulation (transcutaneous) in neurogenic OAB, with most improving; **Stoller** then developed PTNS in pig-tailed monkeys and humans.

Technique

- Sequence: supine, medial malleolus upward → **34-gauge stainless-steel needle ≈3–4 cm (three fingerbreadths) cephalad to the medial malleolus**, between posterior tibial margin and soleus, at **60–90°** → tip near, **not touching**, the nerve → grounding pad on the same leg → **9 V stimulator, 0–10 mA, pulse width 200 μs, 20 Hz** → confirm position → stimulate at tolerable intensity → **12 sessions × 30 min, 1–3 per week** (Fig. 115.10).
- Confirmation pair: **efferent (motor) = great-toe flexion or fanning; afferent (sensory) = radiating tickling of the sole**.

CLINICAL RESULTS

- Mostly OAB. **Subjective success** (QoL questionnaires) = improved QoL or willingness to continue; **objective success** (frequency-volume chart [FVC]) = **≥50% fall in urgency or urgency urinary incontinence (UUI) and ≥25% fall in day or night frequency**.
- **Urodynamics conflict:** acute DO suppression failed in neurological patients and was rare in others, though one series reported elimination in most; **cystometric capacity rose consistently**.
- **SUMiT trial (landmark):** multicentre double-blind RCT, **12 weeks PTNS vs validated sham, n = 220** — the **first validated sham**. Primary endpoint: 7-point self-reported **global response assessment (GRA)**, success = moderate or marked improvement: **55% vs 21%**. FVC improved in frequency, nocturia, moderate-severe urgency voids and UUI. A second sham-controlled RCT agreed.

- **No standard regimen**; most objective data support **12 weeks**; shorter regimens may work but be less durable.
- **Relapse is likely** → maintenance or tapering keeps the brain "remembering" LUT control. Maintenance sustained results at 2 years.
- **SUMiT extension**: 12-week responders on a **14-week taper** then personal schedule (median **1.1 treatments per month**): at **3 years 77% relapse-free**; voids 12.0 → 8.7 per day; UUI 3.3 → 0.3 per day; QoL sustained.
- **Children**: when bladder training, pelvic floor therapy and antimuscarinics fail, PTNS is well tolerated and improves most; pain scores low.

PERCUTANEOUS TIBIAL NERVE STIMULATION COMPARED TO OTHER TREATMENTS

- **PTNS vs tolterodine** (two RCTs, non-neurogenic OAB): both improved at 3 months, **no difference** in QoL, urgency or UUI; **fewer side effects with PTNS**. Other antimuscarinics similar.
- **Cochrane review** (peripheral neuromodulation vs antimuscarinics): subjective improvement favoured stimulation, significant **only for PTNS**, without significant differences in frequency, nocturia, UUI or QoL. Conclusion: **antimuscarinics remain established**; stimulation may be better in **antimuscarinic-refractory** patients.

PROGNOSTIC FACTORS FOR PERCUTANEOUS TIBIAL NERVE STIMULATION

- Few data. Better outcome in patients **without DO** on urodynamics.
- Most clinical parameters non-predictive; only **low Short Form-36 (SF-36) Mental Component Summary** predicted failure — worse disease-specific QoL despite equal severity → **mental-health questionnaires may aid selection**.

IMPLANTABLE TIBIAL NERVE STIMULATORS

- **Rationale**: tapering needs repeated visits (logistics, travel burden), on-demand PTNS is impossible, and surface electrodes are less effective because of **skin impedance**.
- **Not all PTNS responders respond to the implant** — as with sacral stimulation.

IMPLANTABLE TIBIAL NERVE STIMULATORS

Device	Design and placement	Power	Outcome and safety
Urgent-SQ (first; 2006 feasibility; Fig. 115.11)	Subcutaneous, ≈5 cm above medial malleolus , nerve not exposed; activated intraoperatively to confirm position	External stimulator	Half to most met ≥50% reduction in incontinence episodes or voids; AEs: UTI, transient walking difficulty, radiating sensations; no infection, erosion or dislocation. Safe at 9 and 18 years , but most stopped use (lost efficacy, defective external stimulator)
BlueWind (Fig. 115.12)	Placed as Urgent-SQ; 0.6 g, 25 × 3.4 mm	External control unit	Improved frequency and UUI, some patients dry; success sustained at 3 years; AEs: implant-site pain, suspected infection, wound complications
eCoin (investigational)	Nickel-sized, 23.3 × 2.4 mm; in-office, local anaesthesia; template from medial malleolus; 3-cm vertical incision 5–10 mm cephalad to target; suprafascial pocket over nerve	Primary (non-rechargeable) battery	Symptom relief in most, some fully continent; battery replacement feasible with sustained efficacy

Device	Design and placement	Power	Outcome and safety
Protect-PNS (investigational)	Percutaneous; retrograde from medial malleolus ; miniature pulse generator + quadripolar array	Wireless energy	First small series: UUI fell, no complications

- Pattern: all sit over the tibial nerve just above the medial malleolus; they differ in **power source** (external, primary battery, wireless). Implants are **well tolerated and safe without long-term complications**; the trap is assuming PTNS response predicts implant response.

CONCLUSIONS

- PTNS modulates storage and voiding: **subjective improvement ≈60%, FVC improvement 47–56%, placebo effect ≈21%** (possibly higher in children), **53–77% still improved at 2–3 years with maintenance**.
- **Minimally invasive, inexpensive, but time-consuming** → implants under investigation.

KEY POINTS: PERCUTANEOUS TIBIAL NERVE STIMULATION

- PTNS resembles Chinese acupuncture.
- First urological tibial nerve stimulation report: **1983**.
- **SUmIT trial: PTNS superior to sham**.
- First feasibility report of a tibial implant: **2006**.
- Modern tibial implants are **nearing clinical introduction**.

Concept — PTNS is **proven against sham but not durable without maintenance**; every later development — tapering, TTNS, implants — is an answer to the maintenance burden, not to efficacy.

SACRAL NERVE STIMULATION

- SNS is an **effective third-line therapy for OAB**.

CURRENT INDICATIONS AND OUTCOMES FOR SACRAL NERVE STIMULATION

GUIDELINE POSITION OF SACRAL NERVE STIMULATION IN IDIOPATHIC OVERACTIVE BLADDER

Line	American Urological Association / Society of Urodynamics, Female Pelvic Medicine and Urogenital Reconstruction (AUA/SUFU)	European Association of Urology (EAU)
First	Lifestyle intervention and pelvic floor exercises	
Second	Pharmacotherapy	
Third	SNS when first and second line fail — positions identical	

SACRAL NEUROSTIMULATOR MILESTONES

Year	Event
1994	InterStim (Medtronic), first non-rechargeable sacral neurostimulator: Conformité Européenne (CE) mark for functional pelvic, LUT and bowel disorders
1997	FDA approval for UUI
1999	FDA approval for urgency-frequency and NOUR
2004	First rechargeable battery for spinal cord stimulation (pain)
2006	InterStim II approved (non-rechargeable, voltage-driven)

Year	Event
2019	First rechargeable bladder neurostimulator (Axonics); FDA approval of Axonics and Medtronic rechargeable devices
2021	Both companies introduce non-rechargeable devices lasting ≥10 years

- Two US manufacturers (**Axonics, Medtronic**), both offering **MRI-compatible rechargeable and non-rechargeable** systems of **comparable effectiveness**; they differ in user interface and recharging frequency.

OVERACTIVE BLADDER WITH OR WITHOUT URGENCY URINARY INCONTINENCE

- **International Continence Society (ICS):** OAB with incontinence = involuntary leakage **preceded or accompanied by urgency** (sudden compelling desire to void, difficult to defer).
- **Upper limit of normal = 7 voids per 24 h** on 1.5–2 L intake; **OAB without incontinence = ≥8 voids per 24 h with urgency**. The definition does not exclude **excess fluid intake** → **frequency-volume diary before any OAB treatment**.
- **Exclude:** UTI, underactive bladder with significant PVR, calculi, bladder cancer.
- Urodynamic DO may or may not be present; **DO does not predict SNS success**.
- **Common success definition: ≥50% symptom improvement during a 1–2-week test**; definitions vary across studies.

RANDOMISED TRIALS OF SACRAL NERVE STIMULATION IN OVERACTIVE BLADDER

Trial	Comparator and population	Endpoint	Verdict
InSite	Standard medical treatment (SMT); n = 147, mild–moderate OAB	Success at 6 months, intention to treat	SNS 61% vs SMT 42%
Earlier RCT	SMT	Voiding and incontinence at 6 months	SNS significantly better; many fully continent
ROSETTA (6 months)	OnabotulinumtoxinA (BTX) 200 U ; n = 381 women, refractory UUI	Change in daily UUI episodes	BTX –3.9 vs SNS –3.3 — BTX greater
ROSETTA (2 years)	As above	UUI episodes; satisfaction; complications	Equal, sustained UUI reduction ; satisfaction higher with BTX; recurrent UTI 24% vs 10% (P < 0.01); 6% needed clean intermittent catheterisation (CIC) after second injection; SNS revision 3%, removal 9%

- **ROSETTA's 200 U is twice the registered idiopathic OAB dose:** off-label before registration, now not used because of more **retention and UTI**. **Registered dose = 100 U**, superior to placebo at 4 months.
- ROSETTA's lower CIC and SNS revision and removal rates were attributed to **standardised lead placement and strict response criteria**.
- Prospective cohorts at ≥5 years: about half keep fewer incontinence episodes and less pad use.

NONOBSTRUCTIVE URINARY RETENTION

- **NOUR:** non-painful bladder with **chronically high PVR** needing regular drainage; accepted PVR debated, often **≥100 mL**. UAB is usually also **hyposensitive with capacity >500 mL**.
- **Measured PVR is the only objective marker** of success or failure.
- **RCT, SNS vs continued CIC:** at 6 months most implanted patients had stopped catheterising, **none of the controls**. Men respond similarly; **younger men respond better**.

- **Predictors after two-stage SNS:** women with a **psychiatric history (including post-traumatic stress disorder)** → higher odds; men with **previous transurethral resection of the prostate or bladder neck incision** → higher odds; **increasing age lowers odds** in both sexes.

OTHER INDICATIONS

- **Bladder pain syndrome (BPS): not FDA approved.** Offered selectively, especially with **coexisting OAB**, because OAB symptoms give an **objective test-phase measure** where pain is subjective and analgesic-confounded. Small, mostly retrospective data: voided volume rose, pain fell.
- **Neurogenic bladder: not approved**, evidence low. A sham-controlled RCT in first-phase responders favoured SNS-on over SNS-off, but the population was heterogeneous. Systematic review (mostly MS and SCI, mixed OAB and retention): high pooled success, **comparable to non-neurogenic**.

SACRAL NEUROMODULATION: FDA-APPROVED VERSUS NOT APPROVED

FDA approved	Not FDA approved
UUI	Bladder pain syndrome
Urgency–frequency	Pelvic pain syndrome
Chronic NOUR	Neurogenic bladder

Sacral Nerve Stimulation Patient Selection and Implantation Procedure

- Implant = **quadripolar lead + neurostimulator**; success depends mainly on **patient selection**.
- Selected OAB patients could theoretically be implanted directly with **≥90% success**, but practice demands a **test trial for all** — office **percutaneous (peripheral) nerve evaluation (PNE)** or theatre-based **two-stage implant**.
- **Pass criteria:** OAB — **≥50% improvement in ≥1 parameter** (frequency, voided volume, incontinence episodes); retention — **catheterisation frequency reduced ≥50%**, ideally none, fewer UTIs.
- Assessment compares a **3–5-day baseline bladder diary** with a diary kept during the test.

Percutaneous Nerve Evaluation Test

- Sequence: local anaesthesia → insulated needle into the **S3 foramen** by bony landmarks (coccyx, iliac crest, lumbosacral joint) → fluoroscopy optional (anatomy does not always match function) → test stimulation: **sensory responses first**, motor at higher amplitude → temporary lead through needle, needle withdrawn → lead taped to skin, external receiver on belt, wireless external stimulator → mild non-painful perineal sensation → **5–7 days** with bladder diary.
- **Landmarks (Fig. 115.14):** S3 lies **≈9 cm from the coccyx tip and 11 cm from the anal verge, then 1–2 cm lateral**; fluoroscopic "cross-hair" — midline plus lower sacroiliac joints.
- **Positive if ≥1 diary parameter improves ≥50%**, then full SNS (lead + permanent stimulator) in one session. Positive in **≈60% of OAB** in the authors' unit; **not used for NOUR** (higher failure).

SACRAL ROOT RESPONSES DURING TEST STIMULATION

Root	Perineal / anal	Foot and leg
S2	—	Ipsilateral calf and foot external rotation
S3 (target)	Anal "bellows", perineal and genital sensation (tapping or pulling)	Great-toe flexion
S4	Perineal / anal response	No foot or toe response

Staged Implant Procedure

- For patients who **failed or could not tolerate PNE**; introduced because some PNE-positive patients did worse after full implant.
- Sequence: **stage 1** — permanent tined quadripolar lead into S3, placement confirmed fluoroscopically with bellows, ipsilateral toe and genital sensory responses, tines anchoring in the fascia above the sacrum (Figs. 115.13, 115.15) (originally open via 10-cm incision with periosteal fixation; **percutaneous since 2003, ≈30 min**) → temporary extension tunnelled to the flank → test **up to 4 weeks** with programme optimisation → **stage 2 at 2–4 weeks** if ≥50% improvement: neurostimulator in the **buttock**, extension removed. Failure → lead and extension removed.
- **Prophylaxis**: meticillin-resistant *Staphylococcus aureus* (MRSA) cover suggested; authors use intravenous cefazolin + metronidazole.
- **ICS consensus**: **S3 preferred**, S4 sometimes; aim for motor and/or sensory responses on **all four contacts at low amplitude**, though fewer contacts or higher voltage can succeed.

PERCUTANEOUS NERVE EVALUATION VERSUS STAGED IMPLANT

	PNE	Staged implant
Setting	Office, local anaesthesia, no incision	Theatre; local, intravenous sedation or general
Lead	Temporary, prone to dislocation → false test failure	Permanent tined lead , more precise; tunnelling lowers infection
Test duration	5–7 days	Up to 4 weeks , programme optimised
If positive	Full implant in one session	Stage 2 at 2–4 weeks: generator only
Conversion to implant	Lower	Higher
NOUR	Not used	Used

- **Programming**: pulse width **210 μs**, **10–16 Hz**, amplitude at threshold of urogenital or anal sensation; best tolerable sensation at the **lowest amplitude**; unipolar or bipolar.
- **Reprogramming** ≈3 times in year 1, then ≈0.8 per year. **Battery life**: **non-rechargeable ≥10 years**, **rechargeable ≥15 years**.

Additional Perspectives of SNS

Constant Versus Intermittent Stimulation

- SNS is continuous, whereas PTNS works with 30 min every 1–2 weeks → intermittent SNS was tested: **cyclic (16 s on / 8 s off) and 8 h on / 16 h off both equal to continuous**.
- **Closed-loop stimulation**, triggered just before OAB symptoms, is a future option.

Rechargeable Neurostimulators

- Rechargeables make battery-sparing protocols **less urgent**. Two-year OAB results after **direct full implantation without a test**: high success; all patients able to recharge; claimed life **>15 years**.

Other Issues

- **Bilateral SNS is not superior to unilateral** (RCT) → not standard practice: no proven benefit, higher cost.
- **MRI**: all recent Axonics and Medtronic devices have **conditional approval for whole-body MRI at 1.5 and 3 T**. At replacement, **remove all previous components** to guarantee MRI safety.

KEY POINTS: SACRAL NERVE STIMULATION

- SNS is an **established third-line option for non-neurogenic bladder dysfunction**; FDA-approved for **idiopathic OAB and NOUR**, not for bladder pain, pelvic pain syndrome or neurogenic bladder.
- Sacral neurostimulators are **1.5-T and 3-T MRI safe**, rechargeable or non-rechargeable, lasting **at least 10 years**.
- SNS and non-invasive neuromodulation in storage disorders are thought to **activate afferents that modulate forebrain awareness and alertness**.

Concept — SNS is governed by **selection through a test phase**: the objective diary change ($\geq 50\%$, or catheterisation for retention) decides implantation, and every technical refinement — *tined leads, staging, S3 targeting* — exists to make that test predict the permanent result.